

Krishna Ambekar

AI/ML Engineer | Data Scientist | M.Eng., Germany

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Summary | AI/ML Engineer & Data Scientist with 2.5+ years of hands-on experience developing and deploying **end-to-end AI and machine learning solutions** across Generative AI, Computer Vision, NLP, and predictive analytics. Experienced in building **production-ready AI systems**, including Agentic RAG applications, deep learning models, multimodal retrieval pipelines, and MLOps-enabled ML services, using Python, PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, FastAPI, Docker, and AWS. Passionate about building **reliable, scalable AI solutions** that solve real-world business challenges

SKILLS, INTERESTS & OTHERS

- **Programming:** Python, SQL
- **Frameworks:** PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Streamlit, Flask, FastAPI
- **LLMs & NLP:** Prompt Engineering, RAG, LangChain, LangGraph, Hugging Face, BERT, Word2Vec, NLTK, Vector DB
- **MLOps & Deployment:** MLflow, Airflow, Weights & Biases, Docker, AWS (EC2, S3, SageMaker, Lambda, Bedrock), CI/CD Pipelines
- **Computer Vision:** OpenCV, YOLOv8, ONNX, TensorRT, Grad-CAM
- **Soft Skills:** Problem-Solving, Independent & Collaborative Work, Communication
- **Languages:** English (Fluent), German (B2)

WORK EXPERIENCE

Master Thesis Student

08.2025 – 06.2026

VITRONIC Machine Vision GmbH, Wiesbaden, Germany

Thesis title: Vehicle Re-Identification using Deep Features [\[Report\]](#)

- Designed and trained a **Siamese** architecture with ResNet (CNN) and Vision Transformer (ViT) backbones in **PyTorch** to learn discriminative vehicle embeddings.
- Built a real-time **vehicle retrieval** and **pairwise classification** on NVIDIA A100 GPUs, optimized for low-latency inference
- Curated and preprocessed three distinct vehicle dataset with annotation, augmentation, and class balancing to improve model robustness and generalization across varying camera viewpoints and conditions
- Applied **Grad-CAM** and **occlusion** analysis to validate attention maps; visualized embedding separation using **t-SNE** projections
- Achieved **94% pairwise verification** accuracy and improved **Rank-1** by **12%** over baseline using **mAP** and **CMC** evaluation
- Implemented **MLOps** workflows with Weights & Biases; benchmarked FP16/INT8 inference with **PyTorch**, **ONNX**, and **TensorRT**

Internship

01.2025 – 06.2025

Schaeffler Technologies AG & Co. KG, Herzogenaurach, Germany

- Integrated Intel RealSense camera with **YOLOv8** for realtime object detection and human-robot safety at **30-40 FPS inference**
- Built an **AI scouting pipeline** using Schaeffler's internal OpenAI API with structured prompt engineering to auto-extract company fields from **GlassDollar** reports into production-ready JSON, reducing manual data collection effort
- Designed a multi-turn agentic enrichment system using OpenAI function calling for web search and missing data completion
- Built and tested a **RAG chatbot** using **LangChain**, **FAISS**, and **HF** embeddings for semantic product and supplier search; evaluating chunking strategies, re-ranking, and retrieval configurations to optimise LLM context relevance
- Packaged the complete PDF-to-PowerPoint LLM pipeline as a **.exe** for non-technical users to run the full scouting workflow

Research Assistant

03.2024 – 12.2024

CARISSMA- Institute for Vehicle Safety Research, Ingolstadt, Germany

- Prepared and annotated image datasets using **Labellmg** to support supervised learning for object detection models.
- Applied **data augmentation** to expand training samples and improve robustness to lighting and viewpoint conditions.
- Designed **SQL** queries using JOINS, aggregations, and filtering to extract, clean, and transform multi-source telemetry data
- Performed **exploratory data analysis (EDA)** on 28 sensor features and engineered 40+ features including rolling statistics, lagged temporal features, and physics-based terms to predict real-time power in electric vehicles (BMW i3)
- Trained and evaluated **Random Forest**, **XGBoost**, and **CatBoost** regression models with preprocessing pipeline (outlier removal, StandardScaler), achieving best **R² of 0.82 with CatBoost** on unseen driving trips

Graduate Engineering Trainee

10.2022 – 09.2023

Mahindra and Mahindra Limited, Nashik, India

- Developed **PFMEAs**, **POS**, **QC-Plans** and supported **electronics** system documentation for the Thar 502 model
- Leveraged **Industrial IoT** sensor data and **TPM** metrics to drive process optimization decisions, improving **operational efficiency by 23%** and reducing **workplace errors by 20%**

EDUCATION

MASTERS IN INTERNATIONAL AUTOMOTIVE ENGINEERING

10.2023 – 06.2026

Technische Hochschule Ingolstadt, Germany

- Faculty of Electrical Engineering and Information Technology
- Focus on: Artificial Intelligence, Machine Learning and Data Science

BACHELOR IN MECHANICAL ENGINEERING

07.2018 – 05.2022

Savitribai Phule Pune University, Pune, India

- GPA: 9.2/10

PROJECTS

ERP & Procurement Intelligence Assistant [GitHub]

- Developed an **Agentic RAG system** for enterprise procurement teams to answer queries spanning structured ERP data (POs, invoices) and unstructured documents (contracts, policy PDFs), eliminating manual cross-referencing
- Architected a **multi-agent system** using **LangGraph**, integrating PostgreSQL, ChromaDB, FastAPI, and Amazon Bedrock/Groq to intelligently route queries across SQL, RAG, and hybrid pipelines - generating context-aware, cited business responses
- Hardened the pipeline with **input guardrails** (blocking off-topic/unsafe queries, resilient **LLM fallbacks** and LangSmith tracing)
- Deployed on **AWS**; achieved **RAGAS** scores of **0.89 Faithfulness** and **0.92 Answer Relevancy** and **<2 s SQL latency**

Tech Stack: Python, LangGraph, LangSmith, FastAPI, ChromaDB, PostgreSQL, AWS (Bedrock, RDS, EC2, Lambda), Groq, RAGAS, Docker

Voice of Customer (VoC) Analytics Platform [GitHub]

- Built an **NLP platform** for businesses to automatically analyse thousands of customer reviews — identifying which specific aspect (price, delivery, quality) drives dissatisfaction and track whether complaints are increasing over time
- Architected a **3-stage** pipeline: fine-tuned DistilBERT for Aspect-Based Sentiment Analysis (selected via **MLflow**-backed comparison with RoBERTa and DeBERTa), BERTopic for complaint discovery, and LangChain + Groq for report generation
- Achieved **0.76 Weighted F1 across 94K reviews** with **<100ms API response time**, auto-discovering **40–60 complaint topics** with zero manual labelling; deployed using FastAPI and AWS (EC2, RDS, S3, Lambda)

Tech Stack: Python, DistilBERT, BERTopic, LangChain, Groq, MLflow, FastAPI, Streamlit, AWS, Docker, HuggingFace Transformers

Financial Transaction Fraud Detection [GitHub]

- Developed a **fraud detection** pipeline on 6.3M financial transactions, addressing **0.13% class imbalance** through feature engineering and business-driven threshold optimisation
- Built and compared Logistic Regression, Random Forest, and XGBoost models using **SMOTE** and **cost-sensitive learning**, achieving **99.76% recall** and **95.96% precision** with Random Forest through business-driven threshold optimization.
- Implemented **SHAP explainability** for reporting; deployed via AWS SageMaker with MLflow and CloudWatch drift monitoring

Tech Stack: Python, Scikit-learn, XGBoost, SHAP, MLflow, FastAPI, AWS (SageMaker, Lambda, CloudWatch, S3), Docker

Drivable Area Detection using Image Segmentation [GitHub]

- Built a **drivable area detection** system for dashcam, performing 3-class pixel-level segmentation with low-latency inference
- Trained a **U-Net architecture** from scratch in PyTorch on BDD100K using an Albumentations pipeline with Grad-CAM explainability; exported to ONNX, achieving a 2.4× CPU speedup
- Achieved **60% mIoU at 30 FPS (GPU)** and deployed on **AWS** with Dockerized Streamlit app, ECR, and GitHub Actions CI/CD

Tech Stack: PyTorch, U-Net, Grad-CAM, ONNX Runtime, Streamlit, AWS (ECR, EC2), Docker, GitHub Actions, OpenCV

Enterprise Knowledge Assistant [GitHub]

Quora Duplicate Question Pair [GitHub]

ACHIEVEMENTS

- Completed Master's degree at a tuition-free public university in Germany
- Participated in HACK|BAY 2025 (Zollhof Hackathon, Nuremberg, Germany) - built an AI Agents solution for the DATEV-sponsored challenge track among 150 hackers

COURSES AND CERTIFICATES

- AWS Certified Machine Learning Engineer Associate [Certificate]
- LLM Engineering: Master AI, Large Language Models & Agents [Certificate]
- PyTorch for Deep Learning with Python [Certificate]